

Name: Jommel Sean B. Quilon
Course, Year & Section: BSIT 4-2

Student Number: 202300950
Date: 19/09/2026

ONLINE ACTIVITY #2

Basic Linux Commands

- *pwd*: Prints the current working directory path (Usage: *pwd*).
- *ls*: Lists files and folders in the directory (Usage: *ls -la*).
- *cd*: Changes the current working directory (Usage: *cd Documents* or *cd ..*).
- *mkdir*: Creates a new directory (Usage: *mkdir projects*).
- *touch*: Creates a new empty file (Usage: *touch notes.txt*).
- *rm*: Deletes files or directories (Usage: *rm file.txt* or *rm -r old_folder*).
- *cp*: Copies files or directories (Usage: *cp file.txt copy.txt*).
- *mv*: Moves or renames files and folders (Usage: *mv old.txt new.txt*).
- *cat*: Displays the full content of a file in the terminal (Usage: *cat readme.md*).
- *grep*: Searches for specific text patterns inside files (Usage: *grep "error" app.log*).

Basic Git Commands

- *git init*: Initializes a brand new local Git repository in the current folder (Usage: *git init*, Common Flag: *--bare*).
- *git restore*: Discards uncommitted changes in files or unstages them (Usage: *git restore file.txt*, Common Flag: *--staged*).
- *git status*: Shows the state of the working directory and staged changes (Usage: *git status*, Common Flag: *-s*).
- *git branch*: Lists, creates, or deletes local branches (Usage: *git branch feature*, Common Flags: *-a*, *-d*).
- *git add*: Adds modified file contents to the staging area for the next commit (Usage: *git add file.py*, Common Flag: *-A*).
- *git switch* / *git checkout*: Switches to another branch or checks out working files (Usage: *git switch main*, Common Flag: *-c*).
- *git commit*: Records staged snapshots permanently into the repository history (Usage: *git commit -m "Initial commit"*, Common Flags: *-m*, *-a*).
- *git merge*: Joins another branch history into your currently checked out branch (Usage: *git merge feature*, Common Flag: *--no-ff*).
- *git log*: Shows commit logs and project history (Usage: *git log*, Common Flags: *--oneline*, *-n 5*).
- *git tag*: Creates reference points or release markers at specific commits (Usage: *git tag -a v1.0 -m "Release"*, Common Flag: *-a*).
- *git diff*: Shows file modifications between the working directory, stage, and commits (Usage: *git diff*, Common Flag: *--staged*).
- *git clone*: Clones a remote repository onto your local system (Usage: *git clone [https://github.com/user/repo.git]* [https://github.com/user/repo.git], Common Flag: *--depth 1*).

Three-Tree Architecture of Git

- **Working Directory**: The local sandbox where you create, edit, and delete your project files directly.
- **Index (Staging Area)**: The middle layer where selected file snapshots are staged using *git add* before committing.
- **HEAD**: The pointer to the latest commit of your current branch inside the repository record.

References & AI Declaration

- References: Git Documentation (<https://git-scm.com/doc>) and Linux Man Pages (<https://man7.org/linux/man-pages/>).

"I have not used any AI or AI-assisted technologies to prepare this work. I simply used website educational information and edited it personally."